KisanHaat: An Integrated Digital Platform for Agricultural Trading and Machinery Rental

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***Abstract*—Agriculture is still a crucial sector of the Indian economy, but it is still struggling with the issues of the reliance on the middlemen, the lack of price transparency, and the inability to obtain modern farm machinery. Though digital projects such as e-NAM and other agritech start-ups have led to improved connectivity in the market, most of them still do not have a single system that will combine crop trading with resource-sharing services. In this paper, it suggests KisanHaat, a hybrid digital system made up of a farm market and a machine. Rental site, where the farmers will be able to sell their goods, rent equipment on demand and directly. This site will come in handy in increasing transparency, minimizing transactional inefficien cycles and making people cultivate on a sustainable basis. The performance of its technical aspect was measured through Google Lighthouse with the consideration given to accessibility, performance and adherence to the existing development of webbed requirements. The level of accessibility was good in these findings but considerable interface stability (CLS: 0.00) and (score: 86), but the loading speed (LCP: 6.0s) should still be improved. On the whole, the results indicate that KisanHaat can be a massive Indian agricultural sector driver, in terms of digital transformation. More accommodating features, including AI-driven market analytics and support multi-lingual, must be recommended to improve the future. to scale it up in the long term and be more inclusive.**

***Index Terms*—Agritech, Digital Marketplace, KisanHaat, e-NAM, Machine Rental, Indian Farmers**

I. Introduction

In India, agricultural activity is the main source of livelihood to close to half of the population. Small and marginal farmers face traditional challenges that include reliance on the middlemen, inconsistent pricing structures, and the lack of access to modern agricultural equipment and machinery despite its importance despite years of operation [1], [2]. Such restrictions do not only cut their income potential, but also limit the large-scale implementation of modern and sustainable farming practices.

The fast pace of digital technology is a unique chance to revolutionize the agricultural industry because it opens chances to establish direct contact between the farmers and the buyers, enhance a better level of transparency in the transactions, and provide an equitable access to the necessary resources. Government-led initiatives like e-NAM have been influential in launching digital trade platforms in agriculture [3] as well as private agritech applications like DeHaat and Ninjacart have contributed to the improved efficiency of the markets by means of digital logistics and analytics-based consultancy services. However, the majority of these efforts are still focused on the trade and supply chain optimization, and the area of farm machinery accessibility and affordability seems to be the missing link.

The solution to this problem is the suggested **KisanHaat** platform that will unite two vital agricultural services in a single digital solution - an online farm produce marketplace and a farm machinery-renting service. Through the marketplace, the farmers will be able to sell their goods directly to consumers or the retailers bypassing the middlemen, and this will mean higher prices. Simultaneously, the rental service allows farmers to receive quality equipment and vehicles like trackers, mowers, and threshers without much difficulty on a temporary or a pay-as-you-use basis. This makes the initial investments less significant and encourages the sharing of the machines, and it also contributes to the general efficiency and productivity in agriculture. Through a combination of these complementary functions, **KisanHaat** aims to establish a complete digital infrastructure to foster economic empowerment and efficiency of operations among farmers. The site is rural-friendly with a simplistic interface, secure payment system, and accessibility in multiple languages. This paper presents a literature review of related literature, design and implementation of the system and the findings of the performance and usability analysis of this system in the following sections of this paper.

II. Related Work

The digital agriculture environment in India has been developed as a combination of policy-based initiatives and agritech entrepreneurial projects aimed at modernizing the agricultural value chain.

A. Government Platforms

The National Agriculture Market (e-NAM) is one of the landmark developments by the government in an attempt to come up with a single digital trading platform in agricultural products. e-NAM enables the discovery of prices in a transparent manner and reduce the role of intermediaries through the integration of Agricultural Produce Market Committee (APMC) mandis [3], [4]. However, issues like the lack of standardized grading practice, poor digital literacy of farmers and slow adoption of the practice by stakeholders still remain to limit its overall efficiency.

B. Private Agritech Startups.

A number of privately run agritech organizations such as Ninjacart, DeHaat, and AgroStar have brought in new forms of operation, both business-to-business (B2B) supply chains network and full-farm management ecosystems [5], [6]. Furthermore, innovations like Fasal and BharatAgri employ Internet of Things (IoT) and Artificial Intelligence (AI) applications to facilitate precision agriculture and offer data-driven advisory services [8], [9]. These initiatives, though have contributed to efficiency and decision making in certain sectors; they are usually segment oriented. A holistic approach incorporating both online trading and sharing equipment services is not well-developed yet,–a gap that the suggested platform, the name of which is KisanHaat, is aimed to fill.

III. Methodology

The design of the formation of the platform KisanHaat is grounded on user-based and scalable architecture, which is focused on inclusivity, performance, and reliability. Its primary aim is to establish one digital platform that is capable of effectively managing both purchasing and selling of agricultural products as well as running equipment rentals. The platform has also been designed to be functional even in the rural part of India where in most cases internet connectivity is poor.

A. Architectural Design

The architecture KisanHaat uses is a client-server architecture which is highly modular that keeps the presentation and the server-side logic decoupled. The design is scalable, easy to maintain, and the communication between parts is made smooth using standardized API endpoints

Frontend Design: Light weight web technologies like HTML5, CSS3, and JavaScript are used to develop the user compatibility interface that makes this compatible with other devices even on low rhyme out the low-bandwidth smartphones. It is design oriented. on availability, reactivity and minimal utilisation of data which is why it can be employed by consumers in the rural areas with varying levels. of digital literacy. The layouts of the forms are straightforward, the icons are clear and easy to understand and the navigation pathways are extremely user-friendly, allowing the user to 'add produce, peep in the have access to machines and to conduct business with. ease.

Backend Design: The server-side application is developed by the use of Node.js run-time environment combined with the Express.js framework. Such combination enables non-blocking and asynchronous process of requests required in a platform to handle a large number of people at a given time in an efficient manner 4. Structured routing and middleware configuration as well as the API inclusion in the marketplace and in the rental feature are also supported using Express.js.

Database Management: An embraced hybrid database model is provided in order to provide a balance between the scalability, performance and data protection:

item MongoDB: It is the primary No SQL database utilized to store active and non-relational data such as product listing, record of user activities, transactional information and equipment rental history etc. [13]. item SQLite: This is a small encrypted local database, which is used to store sensitive information, such as user authentication tokens and active session tokens. This type of separation of confidentia information and the main one. enhances data and information security and reliability, storage and retrieval respectively.

B. System Flow

Fig. 1 demonstrates the end-to-end operation model of the KisanHaat platform, where the user onboarding process through to the confirmation of the transaction occurs. The flow is separated into five major stages:

1) **User Registration and Authentication:** Data is encrypted and is being verified by using validated credentials, farmers and buyers register to the website at this stage.
2) **Profile Setup:** The users create their profiles based on their respective roles, which are farmer, buyer, or equipment owner.
3) **Listing of Products and Machinery:** Farmers can upload relevant details and images of crops or available machinery for sale or rent.
4) **Marketplace and Rental Operations:** Farmers are able to post pertinent information and pictures of their crops or the machinery they have which they need to sell or rent.
5) **Payment and Confirmation:** Buyers and renters are able to navigate listings of crops and machines, compare and transact to purchase or hire equipment.

IV. System Implementation

The implementation phase will convert the architectural framework into a prototype that works. The development approach will focus on the design of modules, which will allow developing the digital marketplace, testing, and

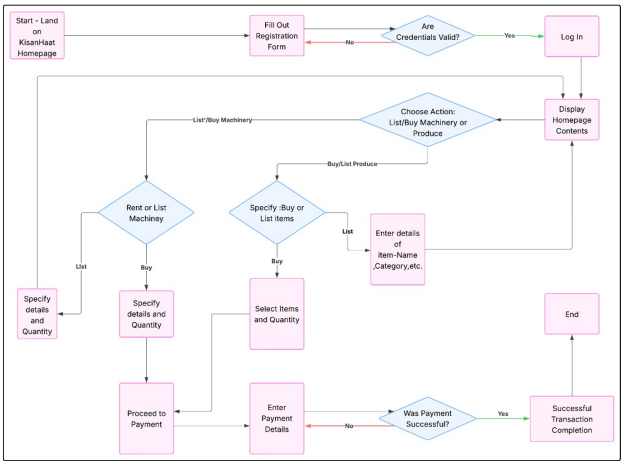

Fig. 1. System Flowchart of the KisanHaat Platform

optimizing it separately, as well as the machinery rental elements.

A. Core Functional Modules

1. User Authentication and Profile Management: The password encryption algorithm is based on the use of bcrypt, and the secure session management algorithm is based on the use of express-session. These functions aid in data security of the user and in avoiding unwanted access. The users can be divided into three major categories such as farmers, buyers, and equipment owners, which contribute to the organization of the platform functions and the personalization of the user interface according to the role of each of the users.

2. Digital Marketplace Module: This module will allow the farmers to post and maintain the listings of agricultural produce and related information details like quantity, price, and location. The customers will be able to browse, filter and buy items online thus they will not have to rely on middlemen. Any money movement is done by encrypted APIs to secure payment gateways such as Stripe or Razorpay, so that no sensitive payment information should be stored on a local computer [16].

3. Machinery Rental Service Module: This module enables short-term and long-term renting of agricultural machinery like tractor, harvester and irrigation machinery as one of the distinguishing feature of KisanHaat. Owners of the equipment are able to establish availability schedules, rates of rental and areas of service. The back end is also used to track real-time inventory movements to prevent multiple reservations and automated alerts notifies the users of rental expiry and renewal availability to encourage openness and responsibility among the participants.

B. Security Implementation

The platform incorporates a multi-level security framework that aims at safeguarding data of users and avoiding cyber attacks.

- **Helmet.js:** This is used to set secure HTTP headers to protect against many common web attacks (like clickjacking and cross-site scripting). item express-validator: Authenticates and purifies user input to avoid script-based and code injection.
- **Data Encryption:** sensitive credentials and tokens of payment are encrypted before being stored with the help of the cryptography functions based on bcrypt and AES.
- **Role-based access:** Role-based authorization is used so that, it makes sure that the users, who can be farmers, buyers, or renters, have access to functionalities only what their respective role on the website.

C. System Evaluation

To check the efficiency and reliability of the prototype, a series of tests such as performance, usability, and security were conducted.

- **Performance Testing:** Apache JMeter was used to model the concurrent user load and test system responsiveness, throughput, and scalability with a load [18].
- **Security Testing:** Static and dynamic testing (SAST and DAST) have been applied based on the OWASP Top 10 guidelines to identify and address possible security vulnerabilities [19].
- **Usability Testing:** The System Usability Scale (SUS) was used to determine the general user satisfaction, accessibility, and ease of navigation, especially to a first-time user [20].
- **Lighthouse Evaluation:** Google Lighthouse audits were performed in-order to evaluate accessibility, SEO compliance, and overall performance of the developed prototype which offers measurable insights in the form of real-world usability.

This organised implementation architecture makes KisanHaat work as a safe, convenient and scalable digital ecosystem- highly consistent with both the technological and operational requirements of the dynamic Indian agricultural industry.

V. Results and Evaluation

Results and evaluation section will comment on the discussion of the findings and their conformity to the research findings. The Google Lighthouse auditing tool was used to evaluate the usage and performance of the the KisanHaat platform. The appraisal was done on four grounds, such as performance, accessibility, best practices and the search engine optimization (SEO). The set of these metrics reflects the effectiveness and practicality of the platform, the readiness, in general, to introduce it to the real situation in rural agriculture.

A. Overall Assessment

A summary of the Lighthouse audit outcomes is presented below:

- **Performance:** 60 - this is an average loading and rendering performance that can be improved with specific optimization measures.
- **Accessibility:** 86 - means that the item is highly adherent to the overall demands of accessibility, which will make it usable by farmers with low levels of digital literacy.
- **Best Practices:** 74 - it implies the compliance with modern web development requirements and the code provisions.
- **SEO:** 67 - implies the equitable presence on search engines, but more specifications may enhance discoverability and online presence.

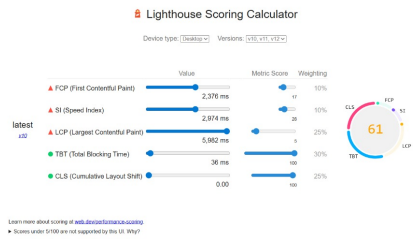

Fig. 2. Overall Lighthouse Evaluation Scores of KisanHaat

B. Detailed Performance Analysis

A comprehensive breakdown of Lighthouse performance metrics is summarized below:

- **First Contentful Paint (FCP):** 2.4s -the duration of the first visible content element, which is an acceptable rendering performance with some possible enhancements.
- **Largest Contentful Paint (LCP):** 6.0s - denotes the delay in the display of the largest visual unit, mainly due to the unoptimized media files and the delay in the response of the server.
- **Speed Index (SI):** 2.9s - indicates average visual loading efficiency, which implies that the majority of content is available within a short time.
- **Total Blocking Time (TBT):** 36ms -represents minimum blocking of main-thread, which is an indication of optimized JavaScript execution and event responsiveness.
- **Cumulative Layout Shift (CLS):** 0.00 -has a high level of interface stability and there is no unplanned layout shifts when loading a page.

C. Evaluation and Interpretation

According to the Lighthouse audit, KisanHaat shows good performance in the areas of accessibility and interface stability thus offering an inclusive and uniform

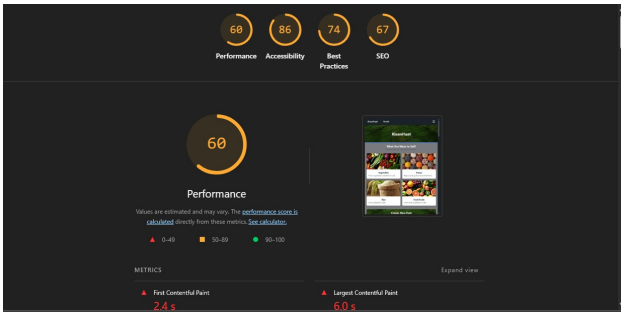

Fig. 3. Detailed Performance Metrics of the KisanHaat Platform

user experience on devices and diverse network strengths. Nevertheless, the fact that the value of the variable **LCP** is quite high indicates that the image processing and server reaction period still require some optimization in order to increase the initial loading rate.

D. Comparative Analysis with Existing Platforms

The proposed platform was compared to such already existing agricultural technology platforms as e-NAM, DeHaat, and Ninjacart. These solutions have been used to enhance agricultural supply chains and digital trading in India but still show some constraints in integration and reachability. Their lot too is cut off to fewer crops.

The platform is called **e-NAM** and it is an initiative of the government to encourage electronic trading in APMC markets [3]. Although it has improved price transparency and integration of digital markets, this is not as inclusive as it should be in areas with a low level of digital literacy due to not having equipment and peer-to-peer functionalities available to it [4].

Data-driven operations in form of the private enterprises like **DeHaat** and **Ninjacart** have developed logistics and market connections [5], [6]. However, they have centralized structures that minimise direct interaction with farmers and do not offer a machinery rental service. Moreover, they cannot be used in low-bandwidth rural locations because they need high-speed connectivity.

Instead, the proposed system named as KisanHaat would be an integrated system that will combine both the crop trading and the machine rental. It is more easily accessible, cheaper and transparent to users in rural trading communities because of its simple interface, multiple language support and low internet connectivity optimization.

The comparative results indicate that the **KisanHaat** application is able to address the functional and accessibility gaps that exist on existing agritech sites. It promotes transparency, reducing the dependency on the intermediaries by merging digital trading and equipment-sharing systems in a single setting and, therefore, facilitates the growth of the community-focused development of agriculture in rural India.

TABLE I
COMPARATIVE EVALUATION OF AGRICULTURAL PLATFORMS

Feature	**e-NAM**	**DeHaat / Ninjacart**	**KisanHaat**
Produce Trading	Yes	Yes	**Yes**
Machinery Rental	No	No	**Yes**
Farmer Interaction	Limited	Limited	**Enabled**
Offline Support	Partial	No	**Yes**
Regional Language	Partial	Limited	**Planned**
Performance (Lighthouse)	N/A	N/A	**60–86**

E. Recommendations for Performance Enhancement

Additionally, we have provided our recommendations on how performance can be improved. The optimization measures proposed to enhance responsiveness of the system and its operational efficiency further are as follows:

1) **Image Optimization:** Adopt responsive image formats (e.g., WebP) and implement lazy loading to reduce bandwidth usage.
2) **Caching and Compression:** Utilize server-side caching and GZIP compression to improve load times and performance during repeat visits.
3) **Code Minification:** Compress CSS, JavaScript, and HTML assets to minimize overall payload size.
4) **Content Delivery Network (CDN):** Employ CDNs for faster global content distribution and reduced latency.
5) **Progressive Web App (PWA) Conversion:** Integrate PWA capabilities to enable offline access and improve the user's mobile experience.

The analysis confirms that **KisanHaat** has a good structural base, security and usability. The system is a scalable, inclusive, and secure solution to the development of the Indian digital agriculture ecosystem despite of the available optimization opportunities due to its performance metrics and design principles.

VI. DISCUSSION AND NOVELTY

KisanHaat is a unique product in the agritech sector because it is both a digital farm produce trade platform and an inbuilt machinery rent platform, which is not common with the existing agricultural technology solutions [5]–[7]. This two-fold framework directly responds to the key areas of concern in Indian agriculture: inefficiencies in market accessibility and the small and marginal farmers lack access to modern farming equipment [22].

Although e-NAM, DeHaat, and Ninjacart, among other eminent platforms, have played a key role in enhancing agricultural trade, logistics, and suggestions processes [3], [5], [6], they mainly work within siloed areas either facilitating the exchange of produce or offering input assistance. Conversely, *KisanHaat* proposes a single ecosystem that enables people to as well as participate in digital produce dealings as well as share or hire farming gears through the use of a peer-to-peer framework.

The inclusion of the machine rental option is one of the biggest innovations as it will allow farmers to use new technology without worrying about the expensive ownership fees. The inclusion facilitates inclusivity, enhances economic stability and sustainable utilization of agricultural resources. Combining digital trading and common access to equipment, , *KisanHaat* is on its way to a decentralized and farmer-oriented model that will encourage fair participation and long-term agriculture development.

A. Limitations

Although it has its potential, the system encounters a number of implementation challenges such as the difference in the digital literacy levels among rural users, the intermittency of the internet connections in the remote areas and a lack of mass implementation of the system in different agro-climatic regions [23]–[25].

B. Future Directions

- Design of multilingual mobile applications and make it more accessible and regionally inclusive [26].
- API Integration of AI modules in dynamic price prediction, crop advisory and yield optimization [27].
- Building partnerships with Farmer Producer Organizations (FPOs) and cooperatives to build trust, scale and adoption of last mile by cooperation [28].

VII. CONCLUSION

This research paper shows that a consolidated digital system, which combines an online farm market with a system to rent farm equipment, is feasible and significant. E-commerce features, combined with resource sharing mechanisms, enable *KisanHaat* to build a scalable, farmer- centric solution that will help to increase income generation, resource utilization, and the overall efficiency of the operation.

Such a two-fold nature of the platform, which allows both the exchange of agricultural goods and equipment sharing, is a great contribution to the agritech environment, as it will resolve accessibility and affordability challenges faced by small and marginal farmers over a long period. In addition, as the planned expansions include AI-driven recommendation engines, predictive data analytics, and multilingual regional integration, it can be stated that KisanHaat will become one of the pillars of digital agriculture change in India [22], [27], [28].

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